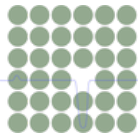




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Paper Presentation

Single-shot Self-supervised Particle Detection in Microscopy

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Single-shot self-supervised object detection in microscopy

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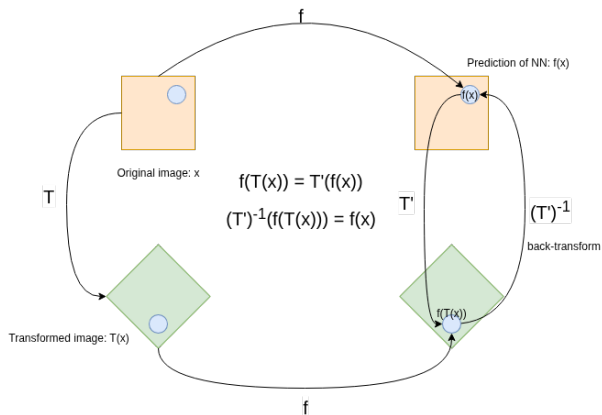
What is LodeSTAR?

- **LodeSTAR:** Localization and **d**etection from **S**ymmetries, **T**ranslations, And **R**otations
- **Goal:** Detect objects in microscopy images without manual annotations
- **Key Innovation:** Self-supervised learning using data transformations
- **Output:** Dense predictions of object positions and confidence weights

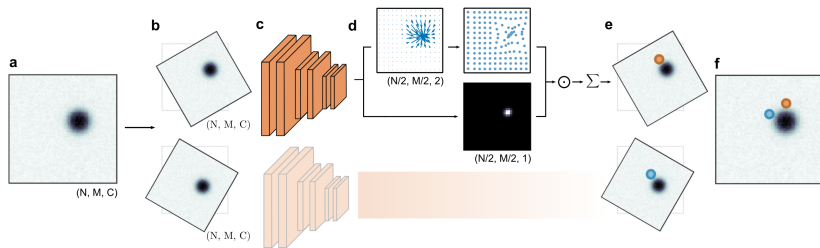
Core Concept: Self-supervised Learning

- **No manual labels required**
- **Data transformations** create supervision signal
- **Consistency across transformations** = learning objective
- **Translation equivariance** ensures robust detection
- **Process:**
 - Transform input image
 - Train model to predict consistent outputs
 - Use consistency as supervision signal

Geometric self-distillation: Equivariance principle



Training



- **Data Augmentation:** Apply k transformations to input image
- **CNN Processing:** Process transformed images through CNN
- **Parameter Optimization:** Optimize parameters to minimize consistency loss

[1]

LodeSTAR Network Architecture

Backbone:

- **Input:** 1-channel grayscale image
- **3× Conv2D (3×3, 32) + ReLU**
- **MaxPool2D (2×2)** after 3rd layer
- **8× Conv2D (3×3, 32) + ReLU**
- **Conv2D (1×1, 3)** → outputs: Δx , Δy , ρ

Core Design Features:

- Uniform 32-channel depth
- Translation-equivariant
- Dense prediction at every pixel
- Lightweight: $\sim 100\text{K}$ parameters
- No fully connected layers

Architecture Summary Table

Layer(s)	Type	Channels	Kernel
0–2	Conv2D + ReLU	$1 \rightarrow 32$	3×3
3	MaxPool2D	–	2×2
4–11	Conv2D + ReLU	$32 \rightarrow 32$	3×3
12	Conv2D	$32 \rightarrow 3$	1×1

Purpose: Regress vector field $(\Delta x, \Delta y)$ and confidence ρ from input image with minimal architecture.

Loss Functions

Within-image Consistency:

$$L_{\text{within}} = \|y_{\text{pred}} - \bar{y}\| \cdot w \quad (1)$$

$$\bar{y} = \sum w_i y_i / \sum w_i \quad (2)$$

- L_{within} : Enforces agreement between all pixel-wise predictions and the consensus (mean) location
- Encourages confident regions to dominate the final prediction

Between-image Equivariance:

$$L_{\text{between}} = \|y_{\text{pred}}^T - T(\bar{y})\| \quad (3)$$

- L_{between} : Ensures predictions remain consistent under transformations
- Encourages the network to learn transformation-equivariant representations

$$\text{Total: } L_{\text{total}} = L_{\text{within}} + L_{\text{between}}$$

Output Processing & Detection

Output Components:

- **Delta X**: X-displacement field ($H/2 \times W/2$)
- **Delta Y**: Y-displacement field ($H/2 \times W/2$)
- **Weights**: Confidence scores ($H/2 \times W/2$)
- **Sigmoid** applied to weights (0-1 range)
- **Coordinate grids** for absolute positions

Detection Process:

1. **Forward pass** through trained model
2. **Extract** δ_x, δ_y , weights
3. **Apply sigmoid** to weights
4. **Generate coordinate grids**
5. **Calculate absolute positions**
6. **Find local maxima** in confidence map
7. **Apply threshold** for final detections

Detection Scoring Metric

Detection Score Calculation:

- The model predicts a weight map (w) and local consistency (b) is calculated from the predictions.
- The final detection score (S) combines these metrics using geometric weighting:

$$S = w^{\alpha} \cdot b^{\beta}$$

- **Parameters:**

- α, β : Weights for confidence (w) vs consistency (b)
- **Cutoff**: Threshold for valid detections

Detection Scoring Metric

- **Local Consistency (b):** Inverse of local variance of feature maps (X) in a 3×3 window:

$$b_{ij}^{-1} = \sum_{c=1}^C \left[\left(\sum_{j'=j-1}^{j+1} \sum_{i'=i-1}^{i+1} \frac{X_{i'j'}^c{}^2}{9} \right) - \left(\sum_{j'=j-1}^{j+1} \sum_{i'=i-1}^{i+1} \frac{X_{i'j'}^c}{9} \right)^2 \right]$$

Summary

- **Single-Shot:** Detects objects in one pass without region proposals.
- **Self-Supervised:** Learns from geometric symmetries, removing the need for manual annotations.
- **Data Efficient:** Works well with limited training data due to equivariance constraints.
- **Lightweight:** Simple CNN architecture ($\sim 100k$ parameters) trainable on standard consumer GPUs.

References

1. B. Midtvedt *et al.*, *Nature Communications* **13**, 7492 (2022).